

Chapter 1: Functions and Graphs

9. Find the domain and range of inverse functions when:

(i) $f(x) = 4x - 3$ (ii) $f(x) = \frac{x}{x-5}$ (iii) $f(x) = \frac{x+2}{x-1}$

(iv) $f(x) = \sqrt{x+2}$ (v) $f(x) = x^2 + 6$ (vi) $f(x) = \frac{2x-1}{x+4}$

Also prove that $f(f^{-1}(x)) = f^{-1}(f(x)) = x$.

7. Show that $f(g(x)) = g(f(x)) = x$, when:
 $f(x) = e^x$ and $g(x) = \ln x$.

Example 18: Draw the graph of $f(x) = e^{-0.5x}$.

Exercise 1.3

1. Draw the graphs of functions.

(i) $f(x) = e^{2x}$

(ii) $g(x) = e^{0.5x}$

(iii) $h(x) = 2 - e^x$

(iv) $h(x) = 1 + e^{-2x}$

(v) $f(x) = \ln(2x)$

(vi) $g(x) = \log(x+1)$

(vii) $h(x) = 3 + \log(x)$

(viii) $f(x) = e^{0.6x}$ and $g(x) = \ln(0.6x)$

Chapter 2: Limit, Continuity and Derivative

In problems 15-18, find the values of m and n so that the given function is continuous.

15. $f(x) = \begin{cases} mx, & x < 4 \\ x^2, & x \geq 4 \end{cases}$

16. $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ m, & x = 2 \end{cases}$

17. $f(x) = \begin{cases} mx, & x < 3 \\ n, & x = 3 \\ -2x + 9, & x > 3 \end{cases}$

18. $f(x) = \begin{cases} mx - n, & x < 1 \\ 5, & x = 1 \\ 2mx + n, & x > 1 \end{cases}$

11. The height above ground of a ball dropped from an initial altitude of 122.5 m is given by

$s(t) = 122.5 - 4.9t^2$, where s is measured in meters and t in seconds.

- What is the instantaneous velocity at $t = \frac{1}{2}$?
- At what time does the ball hit the ground?
- What is the impact velocity?

Example 19: Find $\frac{dy}{dx}$ if $y = [(1 + x^3)\sqrt{x}]$

Find the slope of the tangent line to the curve at the point whose abscissa is given.

14. $y = \frac{4x-1}{x}$, $x = -1$

15. $y = \frac{54}{x^2+1}$, $x = 2$

16. $y = \frac{2x+5}{x+2}$, $x = 1$

17. $y = (2\sqrt{x} + 1)(x^3 - 6)$, $x = 0$

Find the derivative of the given functions w.r.t. x .

11. $y = \sin^{-1}(5x - 1)$

12. $y = 4\cot^{-1}\frac{x}{2}$

13. $y = \frac{\sin^{-1}x}{\sin x}$

14. $y = \frac{\sec^{-1}x}{x}$

15. $y = x\sin^{-1}x + x\cos^{-1}x$

16. $y = \frac{1}{\tan^{-1}x^2}$

Example 36: Find $\frac{dy}{dx}$ if $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$

Example 37: a) Find Δy and dy for $y = 5x^2 + 4x + 1$

b) Compare the values of Δy and dy for $x = 6$, $\Delta x = dx = 0.02$

Example 38: Find the approximated value of $\sqrt{25.4}$.

Use implicit differentiation to find $\frac{dy}{dx}$.

9. $4x^2 + y^2 = 8$

10. $x + xy - y^2 - 20 = 0$

11. $y^4 - y^2 = 10x - 3$

12. $x^3y^2 = 2x^2 + y^2$

13. $xy = \sin x + y$

14. $x + y = \cos xy$

15. $x \sin y - y \cos x = 1$

16. $\sin y = y \cos 2x$

Find $\frac{dy}{dx}$.

17. $y = x^3 e^{5x}$

18. $y = e^{4x}(1 + \ln x)$

19. $y = \frac{e^{2x}}{e^{-2x} + 1}$

20. $y = \ln(e^x + e^{-x})$

21. $y = \ln(x + \sqrt{x^2 + 1})$

22. $y = e^{-3x} \cos x$

Find $\frac{dy}{dx}$ of the parametric functions.

23. $x = t + \frac{1}{t}, y = t + 1$

24. $x = t^2 + \frac{1}{t^2}, y = t - \frac{1}{t}$

25. $x = \frac{\theta^2 - 1}{\theta^2 + 1}, y = \frac{\theta - 1}{\theta + 1}$

26. $x = \sin 2\theta, y = \cos 4\theta$

Find Δy and dy .

27. $y = x^2 + 1$

28. $y = \sin x$

Use the concept of the differential to find the approximated value of the given expressions.

29. $(1.8)^5$

30. $\sqrt{37}$

31. $\sin 31^\circ$

32. $\tan\left(\frac{\pi}{4} + 0.1\right)$

Find the indicated derivative.

15. $y = 4x^7 + x^6 - x^4; \frac{d^4 y}{dx^4}$

16. $y = \frac{2}{x}; \frac{d^5 y}{dx^5}$

17. $f(x) = \cos \pi x; f'''(x)$

18. $f(x) = \frac{1}{\sec(2x+1)}; f^{(4)}(x)$

Example 46: Determine the interval on which the graph of $f(x) = -x^3 + \frac{9}{2}x^2$ is concave upward and the intervals for which the graph is concave downward.

Example 47: Find points of inflection of $f(x) = -x^3 + x^2$

Example 48: Find the critical point and also relative extrema by second derivative test for $f(x) = x^4 - x^2$.

5. Use second derivative test to find the relative extrema of the function.

i. $f(x) = -(-2x - 5)^2$

ii. $f(x) = x^3 + 3x^2 + 3x + 1$

iii. $f(x) = 6x^5 - 10x^2$

iv. $f(x) = x^2 + \frac{1}{x^2}$

v. $f(x) = \cos 3x, [0, 2\pi]$

vi. $f(x) = \cos x + \sin x, [0, 2\pi]$

6. Determine whether the give function has a relative extremum at the indicated points.

i. $f(x) = \cos x \sin x, x = \frac{\pi}{4}$

ii. $f(x) = x \sin x, x = 0$

iii. $f(x) = \tan^2 x, x = \pi$

iv. $f(x) = (1 + \sin x)^3, x = \frac{\pi}{8}$

2. $f(x) = \begin{cases} kx + 1, & x \leq 3 \\ 2 - kx, & x > 3 \end{cases}$ is continuous at 3. What is K?

Chapter 3: Integration

Example 3: Evaluate

$$\int \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}} dx$$

13. $\int \frac{2x+1}{x^2+3} dx$

14. $\int \frac{e^{\tan^{-1} z}}{1+z^2} dz$

15. $\int (x^{\frac{3}{2}} + e^{3x} + x^0) dx$

16. $\int (3x^2 + 2x)(x^3 + x^2 + 9)^5 dx$

17. $\int (5e^{5x} - x^{-3} + 3^{2x}) dx$

18. $\int (z^{\frac{-1}{4}} + \sqrt{3z} + \frac{4}{z} - \frac{1}{e^z}) dz$

Example 5: Prove that:

(i) $\int \sec x \, dx = \ln|\sec x + \tan x| + c$

(ii) $\int \operatorname{cosec} x \, dx = \ln|\operatorname{cosec} x - \cot x| + c$

(iii) $\int \tan x \, dx = -\ln(\cos x) + c = \ln(\sec x) + c$

Example 6: Evaluate $\int \sin^2 x \, dx$

Example 7: Integrate (i) $8\sec 9x - \tan 3x$ (ii) $\cos^2 7x$

3. $\int [1 - 8\operatorname{cosec}^2(2x)] dx$

4. $\int \frac{1}{2} (\operatorname{cosec}^2 x - \operatorname{cosec} x \cot x) dx$

5. $\int \frac{\cos^2 z}{7} dz$

6. $\int (1 + \tan^2 \theta) d\theta$

7. $\int \frac{1 + \cos 4t}{2} dt$

8. $\int \sec^2(5x - 1) dx$

9. $\int (\tan 5x + \cos 7x) dx$

10. $\int (\cot 9y - 3) dy$

Evaluate the integral.

11. $\int (\tan^2 2\theta + \cot^2 2\theta) d\theta$

12. $\int \sin^2\left(\frac{11}{2}y\right) dy$

13. $\int \operatorname{cosec} 11x \tan 11x \, dx$

14. $\int \cos \theta (\tan \theta + \sec \theta) d\theta$

15. $\int \operatorname{cosec}^2\left(\frac{x-1}{3}\right) dx$

16. $\int (\cos x)^{\frac{1}{5}} \sin x \, dx$

17. $\int e^y \sin e^y dy$

18. $\int 9 \tan(x + 7) dx$

Example 9: Integrate: $\int \frac{e^{\tan^{-1} x}}{1 + x^2} dx$

Example 10: Evaluate: $\frac{1}{\sqrt{5-4x-x^2}}$

Example 11: Evaluate: $\int \frac{1}{x^2+4x+5} dx$

Example 12: Evaluate: $\int x(x^2-a^2)^{\frac{3}{2}} dx$

Use suitable substitution, to evaluate the integrals.

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|-----------------------------------|-----------------------------------|---|
| 1. $\int \frac{dx}{x^2+9}$ | 2. $\int \frac{dx}{\sqrt{5-x^2}}$ | 3. $\int (2x+7)(x^2+7x+3)^{\frac{4}{3}} dx$ |
| 4. $\int \frac{x^2}{x^3+1} dx$ | 5. $\int \frac{dy}{y^2+8y+20}$ | 6. $\int \frac{dx}{\sqrt{20-x^2-4x}}$ |
| 7. $\int \frac{x dx}{(4x^2+1)^3}$ | 8. $\int x^4 \sqrt{3x^5-5} dx$ | 9. $\int \frac{2ax+b}{ax^2+bx+c} dx$ |
| 10. $\int \frac{dx}{(1-3x)^2}$ | 11. $\int \frac{z^3}{1+z^4} dz$ | 12. $\int \frac{\cot^{-1}x}{1+x^2} dx$ |

Example 14: Evaluate: (i) $\int x^2 \ln x dx$ (ii) $\int x \tan^{-1}x dx$

Example 15: Apply integration by parts to evaluate:

- (i) $\int \sqrt{a^2-x^2} dx$ (ii) $\int \sqrt{a^2+x^2} dx$

Example 16: Apply integration by parts to evaluate:

$$\int e^{ax} \sin bx dx$$

Evaluate the integrals using integration by parts.

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| 1. $\int \ln x \, dx$ | 2. $\int (\ln x)^2 \, dx$ | 3. $\int \sin(\ln x) \, dx$ |
| 4. $\int x^3 \ln x \, dx$ | 5. $\int y \sin 2y \, dy$ | 6. $\int e^x \cos x \, dx$ |
| 7. $\int x \sec^{-1} x \, dx$ | 8. $\int \ln(2x+3) \, dx$ | 9. $\int x^2 e^x \, dx$ |
| 10. $\int x \cos x \, dx$ | 11. $\int \cos^{-1} x \, dx$ | 12. $\int \tan^{-1} x \, dx$ |
| 13. $\int x \sec^2 x \, dx$ | 14. $\int x^2 \sin^{-1} x \, dx$ | 15. $\int \ln [x + \sqrt{1+x^2}] \, dx$ |
| 16. $\int x^3 e^{x^2} \, dx$ | 17. $\int x^2 \sin x \, dx$ | 18. $\int \frac{\ln x}{\sqrt{x}} \, dx$ |

Example 17: Evaluate: $\int \frac{x^3 - 2x}{x^2 + 3x + 2} \, dx$

Example 18: Evaluate: $\int \frac{x^2 + 2x + 4}{(x+1)^3} \, dx$

Example 19: Evaluate: $\int \frac{3x^2 + 5x + 3}{(x+2)(x^2+1)} \, dx$

Evaluate the integrals using partial fractions.

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| 1. $\int \frac{3x+7}{(x+2)(x+3)} \, dx$ | 2. $\int \frac{4x+9}{x^2+x-12} \, dx$ | 3. $\int \frac{21-8x}{x^2+x-6} \, dx$ |
| 4. $\int \frac{3x+7}{(x+2)^2} \, dx$ | 5. $\int \frac{5x^2-5x+2}{(x+1)(x-1)^2} \, dx$ | 6. $\int \frac{9x^2+3x+29}{(x+1)(x^2+4)} \, dx$ |
| 7. $\int \frac{7x^2+7x+4}{(2x+1)(x^2+x+1)} \, dx$ | 8. $\int \frac{x^3+4x^2+9x+14}{x^2+4x+3} \, dx$ | 9. $\int \frac{1}{x^2-9} \, dx$ |
| 10. $\int \frac{1}{x^3+2x^2+x} \, dx$ | 11. $\int \frac{e^x}{(e^x+1)^2(e^x-2)} \, dx$ | 12. $\int \frac{x}{(x+1)^2(x^2+1)} \, dx$ |

4. Find:

$$\int_1^5 [3f(x) - 2g(x)] dx \text{ if } \int_1^5 f(x) dx = 4 \text{ and } \int_1^5 g(x) dx = 5$$

5. Find:

$$\int_1^4 f(x) dx \text{ if } \int_1^2 f(x) dx = 1 \text{ and } \int_2^4 f(x) dx = 2$$

6. Find:

$$\int_3^{-2} f(x) dx \text{ if } \int_{-2}^1 f(x) dx = 1 \text{ and } \int_1^3 f(x) dx = -5$$

7. Use appropriate formula from geometry to evaluate integrals.

$$(i) \int_{-1}^4 (3-x) dx \quad (ii) \int_0^1 [2 + \sqrt{1-x^2}] dx \quad (iii) \int_2^2 \sqrt{x^3-4} dx$$

Example 25: Evaluate: $\int_0^2 \sqrt{2x^2+1} x dx$

Example 26: Evaluate: $\int_a^b \frac{1}{1-\cos x} dx$ when $a = \frac{\pi}{4}, b = \frac{\pi}{3}$

Example 27: Evaluate: $\int_1^e x \ln x dx$

Evaluate the definite integrals.

1. $\int_{-1}^2 (2x+3) dx$

2. $\int_{-4}^{12} \sqrt{y+4} dy$

3. $\int_0^{\frac{1}{2}} (2x+1)^{-\frac{1}{3}} dx$

4. $\int_0^3 (6x^2 - 4x + 5) dx$

5. $\int_{-2}^1 (12x^5 - 36) dx$

6. $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos \theta d\theta$

7. $\int_0^{\frac{\pi}{4}} \sec^2 2\theta d\theta$

8. $\int_2^4 \frac{x^2+8}{x^2} dx$

9. $\int_{\frac{1}{2}}^{\frac{3}{2}} x - \cos \pi x dx$

10. $\int_1^4 \frac{\cos \sqrt{x}}{2\sqrt{x}} dx$

11. $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin x \cos x dx$

12. $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{1+\cos \theta}{(\theta+\sin \theta)^2} d\theta$

13. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\sec x + \tan x)^2 dx$

14. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 x dx$

15. $\int_1^3 \ln x dx$

16. $\int_2^4 \left(e^{\frac{x}{2}} - e^{\frac{x}{4}} \right) dx$

17. $\int_0^{\frac{\pi}{4}} \frac{1}{1-\sin x} dx$

18. $\int_0^{\frac{\pi}{4}} \tan^{-1} y dy$

19. $\int_0^{\frac{\pi}{2}} \frac{\sin x}{(2+\cos x)(5+\cos x)} dx$

20. $\int_2^5 \frac{1}{x(x+1)} dx$

Example 28: Find the area of the region bounded by the line $2y + x = 8$, the x-axis and, the lines $x = 2$ and $x = 4$.

Example 29: Find the area of the region bounded by graphs of:
 $f(x) = (x - 1)^2$ and $g(x) = 3 - x$

Example 30:

Find the volume of the solid
obtained by rotating the graph $y = x^2$
between $x = 1$ and $x = 2$ about x-axis.

Example 31:

Find the volume of the solid generated when the
region enclosed by $y = \sqrt{x}$, $y = 0$ and $y = 2$
is revolved about the y-axis.

8. Find the area bounded by the curve $y = x^3 + 1$, the x-axis and the line $x = 1$.
9. Find the area of the region enclosed by $x = y^2$ and $y = x - 2$ integrating with respect to y .
10. Find the volume of the solid that is obtained when the region under the curve $y = \sqrt{x}$ over the interval $[1, 4]$ is revolved about the x-axis.

Chapter 4: Differential Equations

Example 4:

Verify that $y = \frac{x^4}{16}$ is a solution of the differential equation $\frac{dy}{dx} - xy^{\frac{1}{2}} = 0$.

3. Verify that the indicated function is a solution of the given differential equation.

(i) $2y' + y = 0$; $y = e^{-\frac{x}{2}}$

(ii) $\frac{dy}{dx} - 2y = e^{3x}$; $y = e^{3x} + 10e^{2x}$

(iii) $y' = 25 + y^2$; $y = 5\tan 5x$

(iv) $y' + y = \sin x$; $y = \frac{1}{2}\sin x - \frac{1}{2}\cos x + 10e^{-x}$

(v) $x^3 dy - 2dx = 0$; $y = -\frac{1}{x^2} + 6$

(vi) $y' - \frac{1}{x}y = 1$; $y = x \ln x, x > 0$

Example 10: Solve $\frac{dy}{dx} = \frac{1}{x \tan y}$

Example 9: Solve $(1 + x)dy - y dx = 0$

Example 11: Solve the initial value problem: $\frac{dy}{dx} = -\frac{x}{y}$, $y(1) = 3$

Example 12: Solve $\frac{dy}{dx} = 2x$ such that $y(2) = 4$.

Example 14: Solve $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$

Example 15: Solve the initial value problem:

$$x \frac{dy}{dx} = y + x e^{\frac{y}{x}} ; y(1) = 1$$

Chapter 5: Kinematics of Motion in a Straight Line

Example 2: A particle is moving such that its position can be determined by the function

$S = t^2 + 2\sin t$. Find its velocity at any time ' t '. Also find its initial velocity and velocity at time $t = \frac{\pi}{3}$ (S is in meters and t is in seconds).

Example 3: The position function of a moving particle is given by $S = \sqrt{t} + \ln(t + 1)$.

Find the velocity and acceleration of the particle at any instant of time ' t '. Also find its velocity and acceleration at $t = 1$ and $t = 4$. Here S is measured in meters and time in seconds.

Example 4: The acceleration of a moving particle at any time ' t ' is given by

$$a = 24t + \cos t$$

Find its velocity and position at any time ' t '. Given that at $t = 0$; $v = 0$ and at $t = \frac{\pi}{2}$; $S = \frac{\pi^3}{2}$.

Example 5:

A stone is projected vertically upwards with a velocity of 10m/sec . Find the height attained by the stone at any time ' t '. When will it attain its maximum height? Also find the maximum height attained by the stone and the total distance travelled when it hits the ground.

Example 8: Find the velocity and acceleration of function $f(t) = 2t^2\hat{i} + 3t^4\hat{j} - t^3\hat{k}$ at $t = 1$.

Example 7: Find the derivative of the vector function $f(t) = 3t^2\hat{i} + 8t\hat{j} - \frac{1}{t^3}\hat{k}$ at $t = 5$.

1. If $\mathbf{r}(t)$ is the position of the particle. Find its domain and the range at given point.

Also find its first and second derivative.

i. $\mathbf{r}(t) = (t + 1)\hat{i} + (t^2 - 1)\hat{j} + 5t\hat{k}, \quad t = 2$

ii. $\mathbf{r}(t) = \frac{t}{t+1}\hat{i} + \frac{1}{t}\hat{j} + t^3\hat{k}, \quad t = -\frac{1}{2}$

iii. $\mathbf{r}(t) = e^t\hat{i} + \frac{2}{9}e^{2t}\hat{j} + 5e^{-t}\hat{k}, \quad t = \ln 2$

iv. $\mathbf{r}(t) = (\cos 2t)\hat{i} + (3\sin 2t)\hat{j} + 5t\hat{k}, \quad t = 0$

2. Find the velocity and acceleration of the function at $t = 0$

i. $\mathbf{r}(t) = (3t + 1)\mathbf{i} + \sqrt{3}t\mathbf{j} + t^2\mathbf{k}$

ii. $\mathbf{r}(t) = \left(\frac{t}{\sqrt{2}}\right)\mathbf{i} + \left(\frac{t}{\sqrt{2}} - 16t^2\right)\mathbf{j} - 2t\mathbf{k}$

iii. $\mathbf{r}(t) = (\ln(t^2 + 1))\mathbf{i} + (\tan^{-1} t)\mathbf{j} + \sqrt{t^2 + 1}\mathbf{k}$

iv. $\mathbf{r}(t) = \frac{4}{9}(t + 1)^{3/2}\mathbf{i} + \frac{4}{9}(1 - t)^{3/2}\mathbf{j} + \frac{1}{3}t\mathbf{k}$

3. Find the scalar valued function in term of the magnitude

i. $\mathbf{r}(t) = (3t - 7)\mathbf{i} + t\mathbf{j} - t^2\mathbf{k}$

ii. $\mathbf{r}(t) = \left(\frac{t}{\sqrt{2}}\right)\mathbf{i} + \left(\frac{t}{\sqrt{2}} + 16t^2\right)\mathbf{j} - 2t\mathbf{k}$

iii. $\mathbf{r}(t) = (\ln(t^2 + 1))\mathbf{i} + (\tan t)\mathbf{j} + \sec t\mathbf{k}$

iv. $\mathbf{r}(t) = \frac{4}{9}(t + 1)^{3/2}\mathbf{i} + \frac{4}{9}(1 - t)^{3/2}\mathbf{j} + \frac{1}{3}t^{3/2}\mathbf{k}$

Chapter 6: Analytical Geometry

Example 1:

Find the value of k if the lines:

$$3x + y - 3 = 0, \quad 5x + ky - 3 = 0, \quad 3x - y - 2 = 0$$

Example 2:

Find (i) circumcenter (ii) centroid and (iii) orthocenter of a triangle ABC with A(0, 0), B(6, 0) and C(0, 6). Also prove that circumcenter, centroid and orthocenter are collinear.

Example 3:

The points $P(-1, 2)$, $Q(3, -2)$ and $R(6, 3)$ are vertices of a triangle PQR. Show that altitudes, right bisectors and medians of the triangle are concurrent.

Example 4:

Find the area of triangle if points $(4, -2)$, $(-2, 4)$ and $(5, 5)$ are vertices of a triangle.

14. Find the area of triangle bounded by the lines:

(i) $4x - 5y + 7 = 0$, $x - 2 = 0$ and $y + 1 = 0$

(ii) $x - 2y - 6 = 0$, $3x - y + 3 = 0$ and $2x + y - 4 = 0$

11. Using the formula of area, find x if the points $P(3, 2)$, $Q(-1, x)$, $R(7, 3)$ are collinear.

12. Vertices of a triangle are $(3, -4)$, $(4, h)$ and $(2, 6)$. Find h if area of the triangle is 10 square units.

Example 5:

Find the straight lines represented by $x^2 - 7xy + 12y^2 = 0$

Example 6:

Find measure of acute angle between lines represented by $6x^2 - xy - y^2 = 0$.

Example 7:

Find a joint equation of straight lines through the origin and perpendicular to the lines represented by $3x^2 + 5xy + 2y^2 = 0$

Example 9:

A triangular park is bounded by three straight roads that meet at points $A(2, 3)$, $B(8, 5)$ and $C(5, 10)$. A city planner needs to determine the area of the park for landscaping purpose. How can he determine the area of the park?

Example 10:

A railway engineer is designing an intersection where two rail tracks meet at a junction. The joint equation representing the tracks is given by $x^2 - 4xy - 12y^2 = 0$. Determine the angle between the individual lines representing the tracks.

Chapter 7: Conic Section

Example 2: Find the centre and radius of the circle $x^2 + y^2 - 6x - 10y + 18 = 0$.

Example 3: Find the equation of the circle passing through the points (1, 2), (2, 3) and (3, 5).

Example 4: Find the equation of the circle passing through the points (1, 0), (0, 1) and having its centre on the line $x - 2y + 3 = 0$.

Example 5: Find the equation of the circle passing through the point (2, 3) and the line $x + y - 4 = 0$ is the tangent to the circle at (3, 1).

Example 6: Find the equation of the circle passing through the point (-1, 2), (3, 2) and touching the line $x - 2y + 1 = 0$.

Example 7: Check whether the line $x + 2y - 3 = 0$ is tangent to circle $x^2 + y^2 = r^2$ or not.

14. The three lines $2x - y + 1 = 0$; $2x + y - 3 = 0$ and $x - 2y + 4 = 0$ touch the circle. Find the centre of the circle. Also find the equation of circle.

Example 8: Find the equation of tangent and normal to the circle $x^2 + y^2 - 4x + 2y - 5 = 0$ at point P(1, 2).

5. Find the equation of the tangent to the circle $36x^2 + 36y^2 - 72y + 11 = 0$ at the point on the circle with abscissa $\frac{1}{2}$.

6. Find the equation of the normal to the circle $x^2 + y^2 - 2x + 2y - 11 = 0$ at the point on the circle with ordinate -4.

7. Circles with equations $x^2 + y^2 - 2y - 3 = 0$ and $x^2 + y^2 - 8x + 4y + 11 = 0$ touch each other externally. Find the equation of the common tangent to the given circles.

Example 11:

Find the equation of parabola with focus at (2, 5) and the equation of directrix is $x + 8 = 0$.

Example 12: Find elements of parabola with equation $y = x^2 - 3x + 7$ and draw its graph.

Example 13: Find elements of parabola with vertex at (2, 3) and focus at (7, 3).

Example 14: Find elements of parabola with focus at (3, -1) and its directrix is $3x - 4y + 1 = 0$.

Example 15: Find elements of parabola with vertex at $(2, 3)$ and equation of directrix is $y = -4$.

1. Find focus, vertex, axis of symmetry, directrix, length of latus rectum, end points of the latus rectum of parabola with the given equations. Also draw the graph of the parabola.

(i) $y^2 = 6x$ (ii) $y^2 = -\frac{3}{2}x$ (iii) $x^2 = 24y$ (iv) $x^2 = -5y$

(v) $y^2 - 2y - 12x - 71 = 0$ (vi) $3x^2 + 42x + y + 149 = 0$

(vii) $4y^2 + 4y = 15 - 32x$ (viii) $9x^2 - 6x = 108y + 26$

2. Find the equation of the parabola in each of the following.

(i) Vertex at origin and focus $(0, -\frac{1}{32})$ (ii) Vertex $(-8, -9)$ and focus $(-\frac{31}{4}, -9)$

(iii) Vertex $(-6, -9)$ and directrix: $x = -\frac{47}{8}$ (iv) Vertex $(5, -1)$ and y-intercept: $-\frac{27}{2}$

(v) Focus $(-\frac{3}{4}, -1)$ and directrix $y = \frac{2}{5}$

(vi) Opens left or right with vertex $(7, 6)$ and passes through $(-11, 9)$.

(vii) Opens up or down and passes through the points $(11, 15)$, $(7, 7)$ and $(4, 22)$.

(viii) Vertex $(10, 0)$; axis of symmetry: $y = 0$; length of latus rectum = 1; $a < 0$

(ix) Vertex $(4, 2)$; axis of symmetry: $x = 4$; length of latus rectum = $\frac{1}{3}$; $a > 0$

(x) Vertex at origin; opens left; distance between focus and vertex is $\frac{1}{8}$ units.

Example 16: Find the equation of the tangent to the parabola $y^2 - 6y - 16x + 25 = 0$ with the slope $1/2$.

Example 17: Find the equations of the tangent and normal to the parabola $y^2 - 6y + 8x - 9 = 0$ at point $P(\frac{1}{4}, -1)$.

Example 18: Find the equation of an ellipse with foci $(-3, 0)$ and $(3, 0)$ and the sum of the distance of any point from the foci is 10.

Example 19: Convert the equation of ellipse $4x^2 + 9y^2 + 8x - 36y + 4 = 0$ in standard form. Also find its elements and draw the graph.

- Find the centre, vertices, co-vertices, foci, eccentricity, length and equation of major axis, length and equation of minor axis, directrices, and length of latus rectums for the following equations of ellipse. Also draw the ellipse in each case

(i) $\frac{x^2}{9} + \frac{y^2}{16} = 1$

(ii) $2x^2 + 3y^2 = 30$

(iii) $\frac{x^2}{49} + \frac{(y-3)^2}{64} = 1$

(iv) $x^2 + 9y^2 + 6x - 90y + 225 = 0$

(v) $16x^2 + 9y^2 - 32x + 36y - 92 = 0$

- From the given information, find the equation of ellipse in each of the following.

✱ (i) Vertices $(-4, 6)$, $(-16, 6)$ and co-vertices $(-10, 2)$, $(-10, 10)$

✱ (ii) Centre at $(-8, 5)$; vertex $(-8, 15)$; Focus $(-8, 5 + \sqrt{5})$

✱ (iii) Eccentricity is $\frac{\sqrt{15}}{4}$; centre $(-5, 5)$; co-vertex $(-8, 5)$

(iv) Eccentricity is $\frac{3}{4}$ and passes through the point $(3, 1)$ with centre at $(0, 0)$; major axis along y-axis.

✱ (v) Focus at $(3, 4)$; directrix $x = 5$ and eccentricity $\frac{2}{3}$.

(vi) Centre at $(1, -1)$; horizontal tangents are $y = 7$ and $y = -9$ and vertical tangents are $x = 6$ and $x = -4$.

Example 20: Find the equations of the tangent to the ellipse $\frac{x^2}{2} + \frac{y^2}{16} = 1$ with slope $\frac{1}{2}$.

Example 21: prove that the line $\sqrt{3}x - 2y + 8 = 0$ is tangent to the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$.

Example 22: Find the equations of tangent and normal to the ellipse $x^2 + 2y^2 - 2x + 4y = 0$ at point $P(0,0)$.

Example 25: Find the equation of hyperbola with centre at $(0, 2)$ and it passes through the points $(1, 1)$ and $(4, 7)$.

Example 26: Find the equation of hyperbola with centre at origin with eccentricity $\frac{3}{2}$ and length of its latus rectum is 3.

Example 27: Find the equation of hyperbola in the standard form with centre at origin and eccentricity $\frac{4}{3}$; also, length of its semi conjugate axis is 3, which is along y - axis.

Example 28: Find the equation of hyperbola with centre at origin and eccentricity $\sqrt{3}$. Its one focus is $(3,0)$ and the directrix is $x = 1$.

1. Find the centre, vertices, co-vertices, foci, eccentricity, length and equation of transverse axis, length and equation of conjugate axis, directrices and length of latus rectums for the given equations of hyperbola. Also sketch the hyperbola in each case.

(i) $\frac{(x-5)^2}{36} - \frac{(y-4)^2}{81} = 1$

(ii) $\frac{(y-10)^2}{10} - \frac{(x-1)^2}{15} = 1$

(iii) $-16x^2 + 9y^2 + 32x + 144y - 16 = 0$

(iv) $x^2 - 4y^2 + 2x - 40y - 135 = 0$

Example 29: Find the equations of tangents to the hyperbola $\frac{x^2}{3} - \frac{y^2}{2} = 1$ with slope $\frac{7}{3}$.

Example 30: Check whether the line $x - y - 1 = 0$ is the tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{3} = 1$

Example 31: Find the equation of tangent and normal to the hyperbola $4x^2 - 9y^2 - 16x - 18y - 29 = 0$ at the point $(5, -1)$.

Chapter 8: Inverse Trigonometric Functions and Their Graphs

Exercise 8.1

Q2 Complete

Q3 (x to xii)

Q4 (iv to x)

Exercise 8.2

Example 7,8,9

Q1 (I, ii, iii, ix)

Q2 (I, ii, vii)

Q3 (i, iii)

Exercise 8.3



Examples 10 to 16

Q1 (iii, iv, ix)

Q3 (iii, iv, vii, x, xi)

Q4 (i, ii, iv)

Q5 (ii, iii, iv, vi, x, xi)

Chapter 9: Solution of Trigonometric Equations

Do All examples of the whole chapter

Chapter 10: Numerical Methods

Do All examples of the whole chapter